

MATH3171 (Mathematical Biology III)
YEAR 2025–26, EPIPHANY TERM

PROBLEM SHEET 5

This problem sheet is due in by **5pm** on **Wednesday 4 February 2026**. You may submit it on paper to my office (MCS 3114) or electronically on Gradescope.

Topics covered: Chapter 9

1. **Scalar Difference Equations** Consider the *Ricker model* originally developed to study populations of fish:

$$u_n = u_{n-1} \exp \left(r - \frac{ru_{n-1}}{K} \right), \quad (1)$$

where $r > 0$ and $K > 0$ are given parameters¹.

- (a) Why is this model more biologically plausible than the logistic map? Hint: Think in terms of permissible solutions and initial conditions.

SOLUTION: One reason the logistic map was poorly behaved is that the right-hand side could become negative for some parameters (r too large) or initial condition choices, which would immediately lead to a solution which is not biologically permissible. Here, the right-hand side is always positive for any value of the parameters or the initial condition, so the population can never become negative.

- (b) What are the equilibria of this model, and what do they correspond to physically?

SOLUTION: The equilibria are given by $u^* = 0$ and $u^* = K$, corresponding to an extinction state, and the population being at some carrying capacity K .

- (c) Linearize eq. (1) about each of the equilibria found above, and determine their stability.

SOLUTION: Linearizing the model about an equilibrium u^* we find that perturbations satisfy,

$$u_{1,n} = \underbrace{\frac{K - ru^*}{K} \exp \left(r - \frac{ru^*}{K} \right)}_{f'(u^*)} u_{1,n-1}.$$

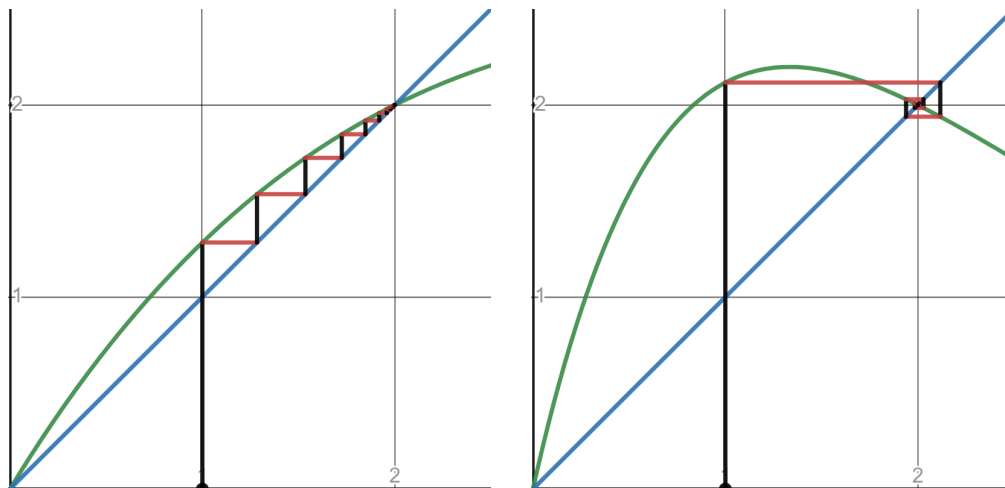
So the extinction equilibrium $u^* = 0$ is stable if $f'(0) = e^r < 1$ which is never true for $r > 0$. Hence this equilibrium is always unstable.

For the positive equilibrium, $u^* = K$, we have that $f'(K) = 1 - r$. Hence for $r > 0$, this equilibrium is stable as long as $r < 2$.

¹Here we write the exponential function as $\exp(x) = e^x$ to avoid confusion with subscripts/superscripts.

- (d) For what values of r do trajectories approach the positive equilibrium monotonically? For what values of r do they approach in an oscillatory way? Draw a cobweb diagram in each case.

SOLUTION: We note that $f'(K) > 0$ for $r < 1$ and $f'(K) < 0$ for $r > 1$. So there is monotonic stability for $0 < r < 1$, and oscillatory stability for $1 < r < 2$.



The two cobweb diagrams are above, with $r = 0.5$ on the left and $r = 1.5$ on the right. See [here](#) for an interactive version.

2. **Systems of Difference Equations** Now let's consider a simple predator-prey system of the form,

$$u_n = ru_{n-1} \exp(-av_{n-1}), \quad (2)$$

$$v_n = u_{n-1} (1 - \exp(-av_{n-1})), \quad (3)$$

where $r > 0$ and $a > 0$ are positive parameters.

- (a) Determine the equilibria of this model, and state any conditions for each equilibrium to be permissible. NB: You can neglect equilibria which only exist for special single values of parameters (e.g. $r = 1$).

SOLUTION: The equilibria are found from the equations,

$$u^* = ru^* \exp(-av^*), \quad v^* = u^* (1 - \exp(-av^*)),$$

from which we see one solution as $u^* = v^* = 0$, which is always permissible. To find others we can divide the first equation by u^* to find,

$$1 = r \exp(-av^*) \implies v^* = \frac{\ln r}{a}.$$

Substituting this into the second equation we find,

$$u^* = \frac{r \ln(r)}{a(r-1)}.$$

The solution above for v^* is only permissible if $r > 1$, hence we need this condition for the second steady state to be permissible at all.

- (b) Determine the stability of each equilibrium. *Hint:* Remember that the trace and determinant of a matrix can be related to the sum and product of its eigenvalues respectively. That is, $\det(\mathbf{J}) = \lambda_1\lambda_2$ and $\text{tr}(\mathbf{J}) = \lambda_1 + \lambda_2$.

SOLUTION: As usual, we first find the Jacobian matrix by computing partial derivatives. This gives us,

$$\mathbf{J}(u^*, v^*) = \begin{pmatrix} re^{-av^*} & -aru^*e^{-av^*} \\ 1 - e^{-av^*} & au^*e^{-av^*} \end{pmatrix}.$$

Evaluation this at the extinction steady state we find,

$$\mathbf{J}(0, 0) = \begin{pmatrix} r & 0 \\ 0 & 0 \end{pmatrix},$$

which has eigenvalues $\lambda = 0, r$. Hence this equilibrium is stable for $r < 1$, and unstable for $r > 1$.

For the second equilibrium, we note that $e^{-av^*} = 1/r$. Using this, we have,

$$\mathbf{J}(u^*, v^*) = \begin{pmatrix} 1 & -\frac{r \ln(r)}{r-1} \\ \frac{r-1}{r} & \frac{\ln r}{(r-1)} \end{pmatrix}.$$

Analyzing the eigenvalues of this matrix in general is difficult. However, using the hint we know that

$$\det(\mathbf{J}) = \frac{\ln(r)}{r-1} + \ln r = \frac{r \ln(r)}{r-1} = \lambda_1\lambda_2.$$

If the eigenvalues lie within the unit circle (that is, the equilibrium is stable), then we must have

$$|\det(\mathbf{J})| = \left| \frac{r \ln(r)}{r-1} \right| = |\lambda_1\lambda_2| \leq |\lambda_1||\lambda_2| < 1.$$

If this were true, however, for $r > 1$ it would imply that $r \ln r < r - 1$, and as these functions are equal at $r = 1$ it would mean this inequality was preserved by taking derivatives, so that $1 + \ln r < 1$, which is not true for $r > 1$. Hence we must have that the positive equilibrium is unstable for all $r > 1$, and so always unstable.

- (c) Is this a biologically realistic model? Justify your answer. Can you suggest ways to improve this model?

SOLUTION: The model seems to be well-behaved for $r < 1$, giving predictions similar to other models we have seen that the populations die out. However, for $r > 1$, the model predicts only unstable solutions, and there

are never any stable coexistence regimes. Related to this, in the absence of predators, the prey population grows without bound. Hence this is likely not a good model.

There are a number of improvements one could make to the model which might help improve stability. A key one would be to add some kind of density-dependent saturation to the prey's response, so that it cannot grow without bound in the absence of predation. In fact, it turns out this leads to a much more realistic model, though one which is harder to analyze without using a computer. See MURRAY, vol. I, Section 3.10 for more details.
